

Discrete optimization

Traveling salesman problem

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Optimization: principles and algorithms



Traveling salesman

We review some classical combinatorial optimization problems and show how they can be modeled as a (mixed) integer linear optimization problem.

In this video, we introduce the traveling salesman problem

Traveling salesman



The traveling salesman problem

- ▶ Consider a network $(\mathcal{N}, \mathcal{A})$ with n nodes representing cities.
- ▶ For any pair (i, j) of cities, the distance c_{ij} between them is known.
- ▶ Find the shortest possible itinerary that starts from the home town of the salesman, visit all other cities, and come back home.



Modeling

Decision variables

$$x_{ij} = \begin{cases} 1 & \text{if } j \text{ is visited just after } i, \\ 0 & \text{otherwise} \end{cases}$$

Objective function

$$\min f(x) = \sum_{(i,j) \in \mathcal{A}}^n c_{ij} x_{ij}$$

Modeling

Constraints

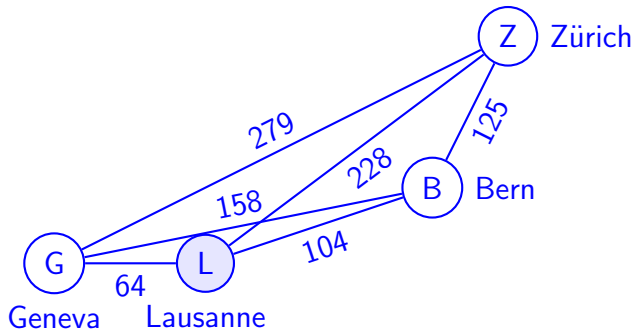
Exactly one successor in the tour

$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij} = 1 \quad \forall i \in \mathcal{N}$$

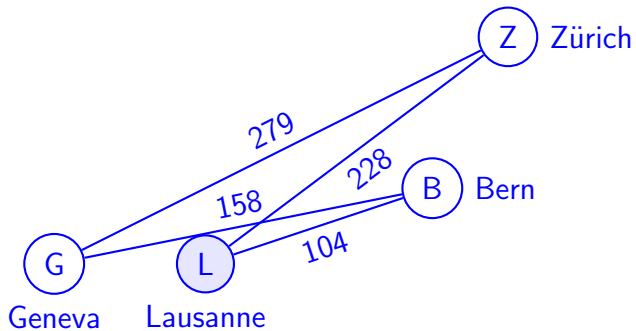
Exactly one predecessor in the tour

$$\sum_{i|(i,j) \in \mathcal{A}} x_{ij} = 1 \quad \forall j \in \mathcal{N}$$

Network



A tour



769km

Subtours

$$x_{LG} = x_{GL} = 1, x_{ZB} = x_{BZ} = 1$$

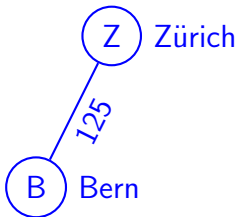
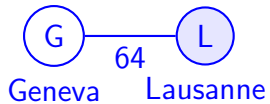
378km

There is exactly one predecessor for each city

There is exactly one successor for each city

There are several ways to eliminate subtours. We present one here.

Subtours



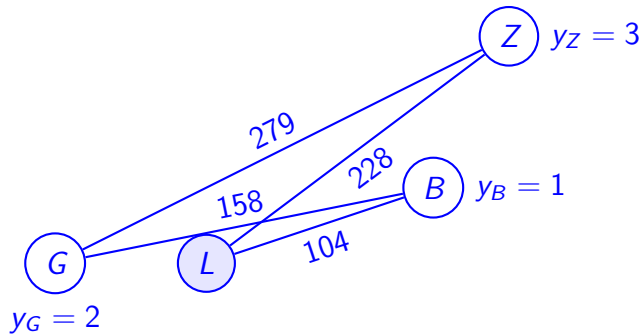
New variables

y_i : position of city i in the tour .

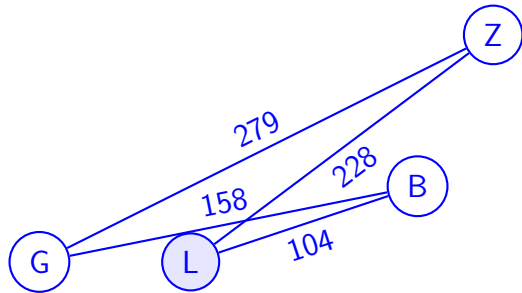
For each i and j different from home:

$$x_{ij} = 1 \implies y_j \geq y_i + 1,$$

A tour



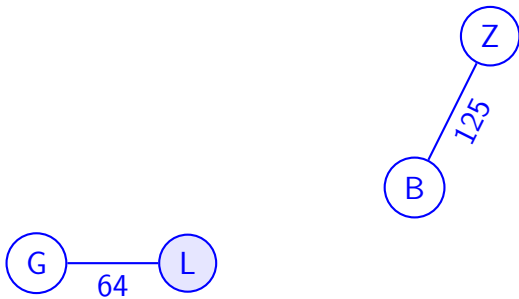
A tour



Subtours

Show that the constraints cannot be verified in subtours not involving home.

A subtour



Additional constraints

$$x_{ij} = 1 \implies y_j \geq y_i + 1.$$

Modeling exercise, using optional constraint see before.

$$x_{ij}(n-1) + y_i - y_j \leq n-2.$$

If $x_{ij}=1$

$$(n-1) + y_i - y_j \leq n-2, \quad y_j \geq y_i + 1$$

If $x_{ij}=0$

$$y_i - y_j \leq n-2$$

Always verified because cities are numbered from 1 to $n-1$

Traveling salesman problem

$$\min_{x \in \mathbb{Z}^{n(n-1)}, y \in \mathbb{Z}^{(n-1)}} \sum_{i=1}^n \sum_{j \neq i} c_{ij} x_{ij}$$

subject to

$$\sum_{j \neq i} x_{ij} = 1 \quad \forall i = 1, \dots, n,$$

$$\sum_{i \neq j} x_{ij} = 1 \quad \forall j = 1, \dots, n,$$

$$x_{ij}(n-1) + y_i - y_j \leq n-2, \quad \forall i = 2, \dots, n, j = 2, \dots, n, i \neq j,$$

$$x_{ij} \in \{0, 1\} \quad \forall i = 1, \dots, n, j = 1, \dots, n, i \neq j,$$

$$y_i \geq 0 \quad \forall i = 2, \dots, n.$$

Summary

The traveling salesman problem is also a classical combinatorial optimization problem.

We have shown that it can be written as an integer optimization problem.